The above condition implies that expectation as of period 1 about C_2 equals C_1. In more general terms if expectations are formed at any period t about a future period t+1, then optimality condition requires that expectation as of period t about C_{t+1} equals C_t, i.e., $C_t = E_t[C_{t+1}]$. Since under rational expectations the actual value of a variable can differ from its expected value only by a random term, this would imply

$$C_{t+1} = E_t[C_{t+1}] + e_{t+1} \quad = C_t + e_{t+1} \qquad \text{.. (8.12)}$$

where e_{t+1} is a random term whose expected value is zero. If you look carefully, equation (8.12) says that consumption from period t to period $t+1$ would remain unchanged, except for an unpredictable random term. This is precisely the conclusion of the random walk hypothesis. The hypothesis says that changes is consumption over time is unpredictable, because they can only change due to the presence of unpredictable random events.

The random walk hypothesis has important implications from the point of view of policy effectiveness. It implies that any government policy to influence consumption (for example, a tax cut) can work only to the extent that it is not anticipated. For example, if the government announces a tax cut policy today, which will be implemented from next year, then consumers with rational expectations will adjust their consumption today itself, so that consumption would remain unchanged when the tax policy actually becomes operative next year.

8.4 CONSUMPTION AND RISKY ASSETS: CAPITAL-ASSET PRICING MODEL

While analysing consumption under uncertainty in the previous Section, we had implicitly assumed that the source of uncertainty is the households' income. In other words, we had assumed that there was no uncertainty as far as the asset return is concerned: the return to asset (r) was treated as given. (In fact, for simplicity we had assumed that it has a value equal to zero). If the return to asset is non-zero, but the exact return is uncertain then the optimality condition of the individual's utility maximization will include an expected return term as shown below:

$$u'(C_t) = E_t[(1 + r_{t+1})u'(C_{t+1})], \qquad \text{... (8.13)}$$

where t is the time period when expectations are formed.

We know from the theorem for statistical expectation (operator E) for two events that A and B $E(AB) = E(A).E(B) + \text{Cov}(AB)$ (where $\text{Cov}(AB)$ is the co-variance between the two events A and B). We can visualize (8.13) as a case of two joint events and write the optimality condition as:

$$u'(C_t) = E_t[(1 + r_{t+1})].E_t[u'(C_{t+1})] + \text{Cov}_t[(1 + r_{t+1}, u'(C_{t+1})] \qquad \text{... (8.14)}$$

Now suppose there are two assets: i) one with a certain return (i.e., risk-free) given by $\bar{i}_{t+1}$, and ii) another with an uncertain risky return with an expected return value equal to $E_t(r_{t+1})$. For the risky asset, if we apply the condition (8.14) we obtain

$$E_t[(1+r_{t+1})] = \frac{u'(C_t) - \text{Cov}_t[(1+r_{t+1}, u'(C_{t+1})]}{E_t[u'(C_{t+1})]},$$

i.e., $$1 + E_t[r_{t+1}] = \frac{u'(C_t) - \text{Cov}_t[(1+r_{t+1}, u'(C_{t+1})]}{E_t[u'(C_{t+1})]}. \quad \text{... (8.15)}$$

For the risk-free asset, the return is certain and therefore is uncorrelated to C_{t+1}. In other words, for the risk-free asset $\text{Cov}_t[(1+\bar{i}_{t+1}, u'(C_{t+1})] = 0$. Hence for the risk-free asset, the condition (8.14) implies that

$$1 + \bar{i}_{t+1} = \frac{u'(C_t)}{E_t[u'(C_{t+1})]}. \quad \text{... (8.16)}$$

Comparing (8.15) and (8.16) we get:

$$E_t[r_{t+1}] = \bar{i}_{t+1} - \frac{\text{Cov}_t[(1+r_{t+1}, u'(C_{t+1})]}{E_t[u'(C_{t+1})]}. \quad \text{... (8.17)}$$

The condition (8.17) gives us a way of determining the optimal (or equilibrium) expected return of a risky asset. Note that a higher value of C_{t+1} implies a lower value of $u'(C_{t+1})$. Therefore a positive co-variance between r_{t+1} and C_{t+1} implies a negative co-variance between r_{t+1} and $u'(C_{t+1})$. Hence (8.17) implies that the higher is the correlation between r_{t+1} and C_{t+1}, the greater is the required expected return on a risky asset compared to the return from the risk-free asset. To put it differently, the greater is the covariance between r_{t+1} and C_{t+1}, the higher is the premium that an asset must offer relative to the risk-free rate. This model of determination of expected return of risky assets is known as the Consumption Capital Asset Pricing Model.

Check Your Progress 2

1. Describe the Random Walk Hypothesis of consumption. What is the implication of this hypothesis from the perspective of government policy?
   ....
   ....
   ....
   ....
2. Explain the Consumption Capital Asset Pricing rule for a risky asset.
   ....
   ....
   ....

8.5 LET US SUM UP

The present unit viewed consumption in a dynamic set up and introduced you to the problem of intertemporal utility maximization. In this context it discussed the issue of consumption and savings when more than one time periods are considered. The Fisherian idea that consumption is an outcome of household's intertemporal optimization exercise has been explained in the unit.

Empirical studies involving cross-section data find that richer households consume a smaller fraction of their income compared to poorer households. However, studies based on time series data show that the proportion of consumption in income has remained more or less the same although household income has increased manifold. This apparent discrepancy has been attempted to be resolved through life-cycle hypothesis and permanent income hypothesis. According to the life cycle hypothesis the short run consumption function shows a declining average propensity to consume (APC) while the long run consumption function exhibits constant APC due to increase in asset base. On the other hand, the permanent income hypothesis explains the discrepancy in terms of permanent income and transitory income.

In the presence of uncertainty actual consumption behaviour may be difficult to predict. In this situation consumption may follow a random pattern. This theory of consumption, known as the Random Walk Hypothesis, has been explained here. Finally, in the presence of uncertainty the price of the risky asset may follow a specific pattern which has been explained in the discussion on the CAPM model.

8.6 ANSWER/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercises 1

1. A change in income will shift the inter-temporal budget constraint. A change in interest rate will affect the slope of the budget constrain. Incorporate the changes in Fig. 8.1 and write your answer.

2. Kuznets found that average propensity to consume remained constant while using aggregate time series data; unlike the Keynesian consumption. In order to explain this difference, two hypotheses are propounded – life cycle hypothesis by Modigliani and permanent income hypothesis by Friedman.

Check Your Progress Exercises 2

1. The random walk hypothesis implies that any government policy to consumption can work only if it is not anticipated.

2. It says that return has to be higher from risky assets. This model is used to analyse market risk premium. See Section 8.4 for further details.

UNIT 9 RAMSEY-CASS-KOOPMANS MODEL*

Structure

9.0 OBJECTIVES

After going through this Unit you should be able to

- explain the Ramsey-Cass-Koopmans problem of optimal growth;
- compare the optimal growth problem of the central planner and that of the decentralized perfectly competitive economy;
- explain the role of government in the optimal growth framework and the corresponding concept of Ricardian Equivalence; and
- if you go through the Mathematical Appendix carefully you should be in a position to solve any standard dynamic optimization exercise.

9.1 INTRODUCTION

In our discussion of household's intertemporal consumption and saving decisions in the previous unit, we considered optimization problems where the time horizon was finite. While this is a valid assumption for an individual, a household (or for that matter, the society as a whole) is generally infinitely lived. In other words, we think of a household consisting of individual members, each of whom live for a finite time period, but new members are born in each time period who are an exact replica of the older members. Thus the household as a dynasty continues to live forever.

An important economic question that is often asked in the context of an infinitely lived household is the following: how should the household allocate its resources between consumption and saving in each time period so that the utility level of each of its members is maximized? In other words, what are the optimal consumption path and optimal saving path for this household? To put is more generally, in an economy consisting of *identical* infinitely lived households, what should be the optimal savings/ accumulation pattern that maximizes the utility of

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each of the households over time? This latter question was first tackled by Frank Ramsey (1928), which was later generalized by Cass (1965) and Koopmans (1965). The model has subsequently been known as the the Ramsey-Cass-Koopmans (RCK) *optimal growth* model.

The Ramsey-Cass-Koopmans optimal growth model essentially involves a dynamic optimization exercise. Here, an economic agent maximizes its objective function defined over a period of time (finite or infinite) subject to some constraints. The constraints are also dynamic in nature in the sense that they relate to values of the variables for different time periods. The technique for solving such dynamic optimization exercises has been specified in the Mathematical Appendix given at the end of this Unit. In the discussion below we will simply state the conditions for optimization (without getting into the underlying technique) and explain the economic intuition behind each of these conditions.

There are two versions of the model: (i) the central planner's problem, and (ii) the decentralized households' problem. You should note that Ramsey introduced his model in 1928 as a central planner's problem in the sense that the planner allocates resources between consumption and saving so as to maximize the utility of the households. Solow model, which is a much simpler growth model compared to Ramsey model, was developed by Robert M. Solow in 1957. Subsequently, the Ramsey model was extended by David Cass and Tjalling Koopmans (separately) in 1965 to include the decentralized households' problem. Keeping in view the similarity of approach in both the versions, they are bracketed together.

9.2 CENTRAL PLANNER'S PROBLEM

In the original Ramsey formulation, the optimal growth problem was postulated as the problem of a social planner who seeks to maximize the welfare of each household subject to the economy's resource constraint at each point of time. The households are all assumed to be identical in terms of preferences and their welfare is represented by an infinite horizon utility function of the form

$$W = \int_0^\infty u(c_t)\exp^{-\rho t}\, dt$$

where c_t is the ***per capita consumption*** in period *t,*

$u(c_t)$ is the associated ***instantaneous utility*** in period *t*, and

ρ is a positive constant term which represents the ***subjective discount rate*** of the household, or its ***rate of time preference***.

Before we proceed further, it is important to explain the presence of the discount factor in the household's welfare function. The positive discount factor reflects the household's preference for present over future. In other words, it implies that a household puts more weight on consumption today vis-à-vis consumption tomorrow. Note that when $t = 0$ (i.e., the initial time period) $\exp^{-\rho t} = 1$. Subsequently when $t = 1$, $\exp^{-\rho t} = \exp^{-\rho}$; when $t = 2$, $\exp^{-\rho t} = \exp^{-2\rho}$ and so on, such that $1 > \exp^{-\rho} > \exp^{-2\rho}$ …….. . Therefore in the infinite-time utility function *W*, current utility at period zero is associated with the highest weight (unity), and each subsequent utility term is associated with lower and lower weights.

The social planner maximizes the integral *W*, but he is constrained by the fact that at each point of time, the economy's total consumption and total investment cannot exceed its total output.[1] To put it formally, $C_t + \frac{dK}{dt} = Y_t$, where C_t is ***aggregate consumption*** in period *t;* $\frac{dK}{dt}$ is the amount of investment in period *t* which augments the capital stock (*K*) of the economy and Y_t is the total output produced in period *t*.

Total population at time *t* is given by L_t, which is assumed to grow at a positive exogenous rate *n*: $\frac{1}{L}\frac{dL}{dt} = n$.

Output at any point of time is produced using the existing capital and labour at that point of time. Technology is represented by a neoclassical production function given by $Y_t = F(K_t, L_t)$. You should note that *F* is continuous, concave and exhibits constant returns to scale (CRS). The CRS property of the production function implies that per capita output (*y*) can be written as the function of the capital-labour ratio (*k*) in the following way:

$$y \equiv \frac{Y}{L} = \frac{F(K,L)}{L} = \frac{L\,F(K/L,1)}{L} = F\left(\frac{K}{L},1\right) \equiv f(k).$$

Moreover, the marginal products of capital and labour can also be written as the following functions of the capital-labour ratio:

$$\frac{\partial F}{\partial K} = f'(k) \ ; \ \frac{\partial F}{\partial L} = f(k) - kf'(k).$$

The continuity and concavity properties of $F(L,K)$ ensure that $f(k)$ is also continuous and concave. Additionally, we assume that $f(0) = 0$; $f'(0) = \infty$; $f'(\infty) = 0$. The first of these assumptions implies that no production is possible with zero capital-labour ratio. The last two assumptions are known as the 'Inada conditions' which state that when the capital-labour ratio tends to zero, the marginal product of capital tends to infinity, while an infinitely high capital-labour ratio implies that the corresponding marginal product of capital approaches zero.

The aggregate resource constraint of the economy is given by $C_t + \frac{dK}{dt} = Y_t$.

[1] For simplicity, we rule out international borrowing here.

Let us divide both sides of the above equation by L. Thus, we can write it as $\frac{C_t}{L_t} + \frac{1}{L_t}\frac{dK}{dt} = \frac{Y_t}{L_t}$

i.e., $c_t + \frac{dk}{dt} + nk_t = f(k_t)$

The above equation denotes the resource constraint faced by the social planner in per capita terms.[2]

The economy starts with a certain given amount of capital stock and population; hence the initial capital-labour ratio is given, denoted by k_0. At every point of time the existing capital and labour stocks are fully employed. Hence, the capital-labour ratio k also denotes the per capita capital stock. While referring to k we shall use these two terms interchangeably.

The complete dynamic optimization problem for the social planner can now be written in terms of the two time dependent variables: (i) per capita consumption (c_t), and per capita capital stock (k_t). The maximization problem before us is

Maximize $W = \int_0^{\infty} u(c_t)\exp^{-\rho t}\, dt$

subject to

$\frac{dk}{dt} = f(k_t) - nk_t - c_t$; k_0 given.

The corresponding Hamiltonian function (H) is given by[3]

$H = u(c_t)\exp^{-\rho t} + \lambda_t\left[f(k_t) - nk_t - c_t\right]$

In the above equation, we have control variable (c_t), state variable (k_t) and co-state variable (λ_t).The first order conditions are obtaind by taking partial derivatives of the Hamiltonian with respect to c_t, λ_t and k_t and and equating it to zero. Thus, we obtain

(ia) $u'(c_t)\exp^{-\rho t} = \lambda_t$

(iia) $\frac{d\lambda}{dt} = -\lambda_t(f'(k_t) - n)$

(iiia) $\frac{dk}{dt} = f(k_t) - nk_t - c_t$

The transversality condition is given by

(iva) $\lim_{t\to\infty} \lambda_t k_t = 0$

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[2] Note that $\frac{dk}{dt} = \frac{1}{L}\frac{dK}{dt} - \frac{K}{L}\left(\frac{1}{L}\frac{dL}{dt}\right) = \frac{1}{L}\frac{dK}{dt} - nk$.

[3] See the Mathematical Appendix for the definitions of the Hamiltonian function, control, state and co-state variables in a dynamic optimization problem.

Notice that λ_t is the shadow price of capital (Please go through the Appendix to this Unit) or the value of capital in utility terms. Hence, condition (ia) states that along the optimal path, the present discounted value of marginal utility from consumption would be equal to the shadow price of capital. The economic significance of this condition would become clear if you note that at any point of time one unit of output can be put to two different uses – one can either consume it or invest it which augments the capital stock. Now optimality condition requires that the returns in terms of utility from these two uses should be equal, which is precisely what condition (ia) states. Conditions (iia) and (iiia) respectively show how the co-state and the state variable changes over time along the optimal path. The fourth condition, known as the ***transversality condition***, implies that as the economy approaches its terminal time (which in this case is infinity), either the value of capital goes to zero (in which case it does not matter for the economy if it leaves a positive capital stock at the end), or, if the value of the capital stock is positive, then the economy uses up all its capital stock (i.e., k goes to zero).

Differentiating (ia) with respect to t and using (iia) to eliminate λ_t, we get the following differential equation: $\frac{dc}{dt} = \left(\frac{u'(c_t)}{-u''(c_t)}\right)\left(f'(k_t - n - \rho\right)$. Noting that the term $\frac{-cu''(c)}{u'(c)} \equiv \sigma$ denotes the elasticity of marginal utility with respect to consumption, the above equation can be written as:

(va) $\frac{dc}{dt} = \left(\frac{c_t}{\sigma}\right)\left(f'(k_t - n - \rho\right)$

Equations (iiia) and (va) together form a system of differential equations which, along with the transversality condition, determine the movements of per capita consumption and per capita capital stock of the economy along the optimal path. The qualitative characterization of this path is shown in the phase diagram given at Fig. 9.3.

The phase diagram traces the lines/curves along which $\frac{dc}{dt} = 0$ and $\frac{dk}{dt} = 0$. The point of intersection of these two lines is called the ***steady state***. A steady state is equivalent to long run equilibrium whereby the values of the variables remains constant over time.

From (va), $\frac{dc}{dt} = 0$ implies either $\frac{c_t}{\sigma} = 0$ or $f'(k_t - n - \rho) = 0$. In the first case, $c = 0$. In the second case, $f'(k) = n + \rho$. Thus, the equilibrium will remain on these two lines. Let us consider equation (iiia). When $\frac{dk}{dt} = 0$, we have $f(k_t) - nk_t - c_t = 0$. On re-arranging terms and ignoing the subscripts (since the values will remain unchanged under equilibrium conditions), we obtain $c = f(k) - nk$. Let us first look into the dynamics of the differential equation

$\frac{dc}{dt} = \left(\frac{c_t}{\sigma}\right)\left(f'(k_t - n - \rho\right)$. Since it can be represented by the lines $c = 0$ and $f'(k) = n + \rho$, we can describe the dynamic of consumption in Fig. 91.

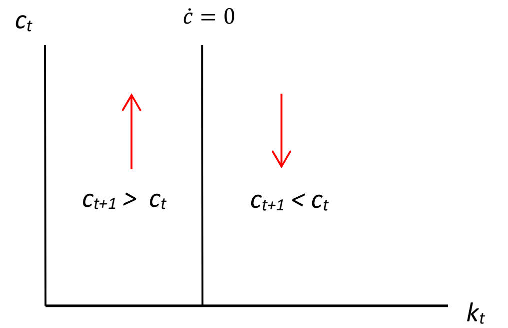

Fig. 9.1: Dynamics of c

In Fig. 9.1, we take per capita capital stock (k) on the x-axis and per capita consumption (c) on the y-axis. The vertical line shows the equilibrium condition where there is no change in per capita consumption over time. Therefore, along this line, we have $\frac{dc}{dt} = \dot{c} = 0$. To the left of this vertical line, we have $\dot{c} > 0$. It implies there will be increase in consumption per capita over time (i.e., $c_{t+1} > c_t$). We have indicated the direction of such change in per capita consumption by an upward-sloping red arrow. To the right of the vertical line, on the other hand, $\dot{c} < 0$, which implies a decrease in per capita consumption over time (thus, $c_{t+1} < c_t$ in this case). If the economy is operating to the right of the vertical line (indicating per capita consumption at a higher than the equilibrium level), there will be decline in per capita consumption over time. We have indicated the direction of such change in per capita consumption by a downward-sloping red arrow in Fig. 9.1.

Now let us discuss how per capita capital stock evolves over time. When $\frac{dk}{dt} = \dot{k} = 0$, we have $c = f(k) - nk$. The equilibrium level $\dot{k} = 0$ is represented by an inverted-U-shaped curve in Fig. 9.2. As you can see from Fig. 9.2, per capita consumption increases with k up to certain leven and declines afterwards. In the region below the curve, $\dot{k} > 0$ (thus, $k_{t+1} > k_t$). We have indicated the direction of such change in per capita capital stock by a right-facing blue arrow. If the economy is oerating in the region above the curve, $\dot{k} < 0$ (thus, $k_{t+1} < k_t$). We have indicated the direction of such change in per capita capital stock by a left-facing blue arrow in Fig. 9.2.

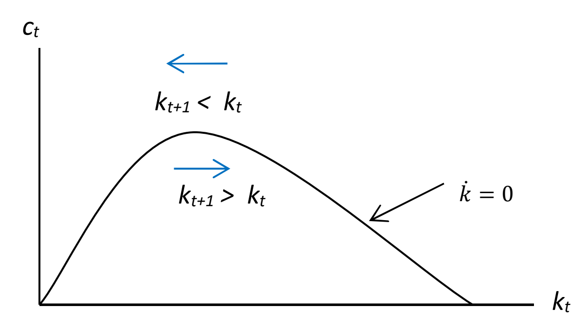

Fig. 9.2: Dynmics of k

Now let us combine Fig. 9.1 and Fig. 9.2. Plotting all these lines and curves in the *c*-*k* plane with *c* along the vertical axis and *k* along the horizontal axis, we get a diagram as shown in Fig. 9.3.

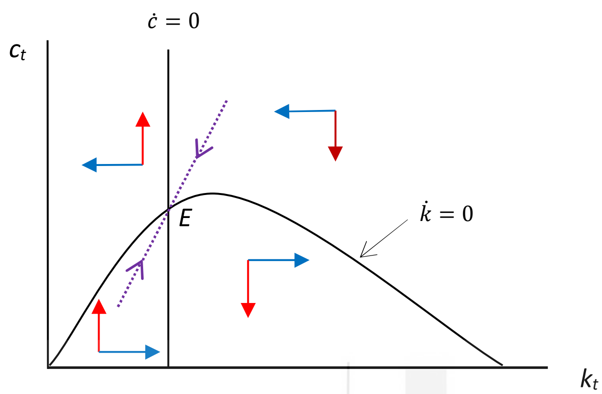

Fig. 9.3: Phase Diagram

The direction of arrows in the phase diagram shows the direction of movements of *c* and *k* when they are not on the $\frac{dc}{dt} = 0$ and $\frac{dk}{dt} = 0$ lines respectively. From Fig. 9.3 it is clear that there exits one steady state, denoted by point E in the diagram, which is associated with positive values of *c* and *k*. It can be shown (though we make no attempt to prove it here) that this steady state point is a 'saddle point' such that there exits a unique path which approaches the steady state point; all the other paths move away from the steady state point. This unique path (called saddle path) is denoted by the dotted purple line in the diagram. This is the only path which satisfies all the first order conditions of optimality (conditions (ia)-(iva)) including the transversality condition. This path is therefore the optimal path which satisfies the social planner's dynamic optimization problem. Given any initial capital-labour ratio, the social planner will choose the initial per capita consumption so as to be on this optimal path, which will eventually take the economy to its long run equilibrium (or steady state) point E. If the economy is operating on any point on the saddle path, it will converge to the saddle point (E). If the economy, on the other hand, is operating at any other point on Fig. 9.3, it will move farther away from the steady state.

At this juncture it is important to point out certain important features of the steady state point E. As was mentioned before, at the steady state the per capita capital stock, *k*, and per capita consumption, *c*, are constant. Also, at this non-zero steady state, the *k*-value and the corresponding *c*-value are defined respectively by the equations: $f'(k) = n + \rho$ and $c = f(k) - nk$. The condition that $f'(k) = n + \rho$ is called the ***modified golden rule***. It states that the steady state value of the capital-labour ratio is such that the marginal product of capital is equal to the sum of the rate of population growth and the rate of time